



2024, Spring.

1. Let $A, B \in M_{m \times n}(F)$. Show that $\text{rank}(A+B) \leq \text{rank } A + \text{rank } B$. 5m

Proof. By definition,

$$\text{rank}(A+B) = \dim \text{Im}(A+B) \text{ and } \text{rank } A = \dim \text{Im } A, \text{ rank } B = \dim \text{Im } B.$$

Given $\alpha \in \text{Im}(A+B)$, $\alpha = (A+B)\beta = A\beta + B\beta \in \text{Im } A + \text{Im } B$ for some $\beta \in F^n$, therefore $\text{Im}(A+B) \subseteq \text{Im } A + \text{Im } B$. Now,

$$\text{rank}(A+B) \leq \dim(\text{Im } A + \text{Im } B) \leq \dim \text{Im } A + \dim \text{Im } B = \text{rank } A + \text{rank } B.$$

2. Suppose that P and Q are positive definite such that $P^2 = Q^2$. Show $P = Q$.

Proof. Since P is Hermitian, there exists an orthonormal basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ such that $P\alpha_i = C_i\alpha_i$ for some $C_i > 0$, for each i .

Let $1 \leq i \leq n$ be fixed. Then, by the assumption,

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$$P^2\alpha_i = C_i^2\alpha_i = Q^2\alpha_i \Rightarrow (Q^2 - C_i^2 I)\alpha_i = (Q - C_i I)(Q + C_i I)\alpha_i = 0.$$

Since Q is positive definite, $(Q - C_i I)\alpha_i = 0$, because: if it is not zero, then $(Q - C_i I)(Q + C_i I)\alpha_i = (Q + C_i I)(Q - C_i I)\alpha_i = 0$,

so $-C_i$ is negative eigenvalue of Q , it contradicts to Q is positive definite. Therefore, for each $1 \leq i \leq n$,

$$P\alpha_i = C_i\alpha_i = Q\alpha_i,$$

and since $\mathcal{B}$ is basis, therefore $P = Q$. $\square$

2024, Spring

3. Let $T: \mathbb{R}^5 \rightarrow \mathbb{R}^5; x \mapsto Ax$ where

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

a). There exists T -invariant subspace $V \subseteq \mathbb{R}^5$ s.t. $\dim V = 4$?

b). "

s.t. $\dim V = 2$, $T|_V$ is diagonalizable?

c). "

$\dim V = 3$, "

Proof.

a). Let $V = \langle e_1, e_2, e_3, e_4 \rangle$. Then,

$$Ae_1 = e_1, Ae_2 = e_1 + e_2, Ae_3 = e_3, Ae_4 = e_3 + e_4$$

are all contained in V , so $T[V] \subseteq V$. (clearly $\dim V = 4$)

b). Let $V = \langle e_1, e_3 \rangle$. Then $Ae_1 = e_1$, $Ae_3 = e_3$ are contained in V , and

$$[T|_V]_{\{e_1, e_3\}} = I_2. \text{ So } T|_V \text{ is diagonalizable.}$$

c). Observe characteristic polynomial of T is $(t-1)^5$.

If such a subspace V exists, then the characteristic polynomial of $T|_V$ is

$(t-1)^3$. Since $T|_V$ is diagonalizable if and only the minimal polynomial

splits and separable, thus the minimal polynomial of $T|_V$ must be $(t-1)$.

That is, $T|_V = I_V$ but it implies that $\ker(T-I)$ has dimension more than or equal 3, it contradicts to

$$\dim \ker(A-I) = \dim \ker \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} = 2.$$

2025, Fall

1. a). Show that $\phi: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p'(0)$ is linear functional.

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b). Find a dual basis corresponding to $\{1, t, \dots, t^n\}$ of $P_n(\mathbb{R})$.

Proof. Let $f(x) = \sum_{i=0}^n a_i x^i$ and $g(x) = \sum_{i=0}^n b_i x^i$ and $C \in \mathbb{R}$ be given.

Then, $\phi(Cf + g) = (Cf + g)'|_{t=0} = C \cdot f'|_{t=0} + g'|_{t=0} = C \cdot \phi(f) + \phi(g)$

by the linearity of differentiation. This shows the a),

To show b), let

$$\mathcal{B}^* := \left\{ \frac{1}{k!} \cdot \phi_k: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p^{(k)}(0) \mid k=0, 1, \dots, n \right\}$$

Then, for each $k=0, 1, \dots, n$,

$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \text{ and } \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$

Similarly a), the elements in $\mathcal{B}^*$ are all linear functional, so the $\mathcal{B}^*$ is dual basis.

2025. Fall

2. Let $A \in M_{\text{max}}(1\mathbb{R})$.

15m: A^n also 연산, $n=2$ 만 보이기 시작

a). For $n \in \mathbb{N}$, $\text{rank } A^n \leq \text{rank } A$.

b). Let $L_A: \mathbb{R}^m \rightarrow \mathbb{R}^m, x \mapsto Ax$.

Show that for $n \geq 2$,

$\text{rank } A^n = \text{rank } A$ implies $\ker L_A \cap \text{Im } L_A = \{0\}$.

Sol) a). Given $x \in \text{Im } A^n$, $x = A^n y$ for some $y \in \mathbb{R}^m$.

So, $x = A^n y = A^{n-1} A y \in \text{Im } A$, for $n \geq 2$. Thus $\text{Im } A^n \subseteq \text{Im } A$, it follows $\text{rank } A^n \leq \text{rank } A$. If $n=1$, clear.

b). Given $x \in \ker A$, $A^n x = A^{n-1} A x = A^{n-1} 0 = 0$, so $x \in \ker A^n$, so $\ker A \subseteq \ker A^n$. Since $\text{rank } A^n = \text{rank } A$, the dimension theorem gives $\text{nullity } A^n = \text{nullity } A$.

Therefore, since $\mathbb{R}^m$ is finite dimensional, so is $\ker A^n$, thus $\ker A = \ker A^n$.

Now, let $x \in \ker L_A \cap \text{Im } L_A$ be given. Then, $Ax = 0$ and $x = Ay$ for some $y \in \mathbb{R}^m$.

Then, $A^{n-1} x = A^n y = 0$, because $Ax = 0$, so $A^{n-2} Ax = 0$, and this implies $y \in \ker A^n$.

Since $\ker A^n = \ker A$, $Ay = 0$, so $x = Ay = 0$, so proved. $\square$

2025. Fall

3. Find all $a, b, c \in \mathbb{R}$ such that

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$$A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix} \text{ is diagonalizable.}$$

Sol. Since the characteristic polynomial of A is regardless of a, b, c ,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the characteristic polynomial, the A is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

$$\text{Calculating gives } = \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$$

therefore, $a=0$, $b \in \mathbb{R}$, $c \in \mathbb{R}$ are all possible (a, b, c) .

2025. Fall.

4. Let $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be the set of all pairs (A, B) such that $\det A = \pm \det B$.

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Show that N is normal and determine the factor group $(G \times G)/N$.

Sol). Let $(A, B) \in N$ and $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

$$\text{and } \det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$$

$$\text{So, since } (A, B) \in N, \det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$$

therefore $(C, D)(A, B)(C, D)^{-1} \in N$. And, to show N is subgroup,

first, $(I, I) \in N$ trivially, so $N \neq \emptyset$. Let $(A, B), (C, D) \in N$ be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

$$\text{satisfies } \det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}.$$

therefore, by the subgroup criterion, N is subgroup, and normal. $\perp$ com.

Define a map

$$\mathcal{Q}: GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \rightarrow \mathbb{R}^{\times}; (A, B) \mapsto (\det A - \det B^{-1})^2$$

Then, $\ker \mathcal{Q} = N$ and $\mathcal{Q}$ is homomorphism, onto $(0, \infty)$, because

the det preserves multiplication, so is $\mathcal{Q}$, and onto is given by: fix $B = I$, and $A = \text{diag}(r, 1, \dots, 1)$ where $r > 0$ is arbitrary.

By the first isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times) \quad \perp \text{ com.} \quad \textcircled{2}$$

Comment: $\mathcal{Q}$ 은 먼저 $\mathbb{R}^{\times}$ 구조 시작하고, hom 임을 보려면 앞 계산을 꼭 해야 함!

12-25. Fall

10m

5. Let G be an order 21 subgroup of $\mathbb{C}^\times$.

a) G is cyclic.

b) For any $g \in G$, $x^{12} = g^{15}$ has 3 distinct solutions in G .

Proof. a) Since $\mathbb{C}$ is abelian, so G is also abelian.

By the Cauchy's theorem, G has order 3 element x and order 7 element y .
Now, since 3 and 7 are relatively prime, $|xy| = 21$, so $G = \langle xy \rangle$.

b) By a), $G \cong \mathbb{Z}/21\mathbb{Z}$. So, we can write the given equation as

$$12x \equiv 15g \pmod{21} \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of g , $\gcd(12, 21) = 3$ divides $15g$, so it has three distinct solutions.

Note: $ax \equiv b \pmod{n}$ has a solution iff $\gcd(a, n) \mid b$.

And in such a case, it has $\gcd(a, n)$ solutions.

2025. Fall.

6. Let $\{a_n\}$ be a bounded sequence in $\mathbb{R}$. Define

22m

$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\}$$

a). Prove that $\{a_n\}^*$ is closed.

b). Show $\limsup_{n \rightarrow \infty} a_n = \sup \{a_n\}^*$

c). Is there a sequence $\{b_n\}$ such that $\{b_n\}^* = [0, 1]$?

Proof a). If $\{a_n\}^*$ is finite, then clearly closed in $\mathbb{R}$.

Assume that $\{a_n\}^*$ is infinite. Since $\{a_n\}$ is bounded, so the set of subsequential limits $\{a_n\}^*$ is bounded set. So it has a limit point.

Let α be a limit point of $\{a_n\}^*$. Then, given $\epsilon \in \mathbb{R}$, there exists a

subsequential limit p of $\{a_n\}$ such that $p \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$.

For $k=1$, choose such a $p_1 \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$, and since p is a subsequential limit, choose $a_{p_1} \in B_{\frac{1}{k}}(p)$, then $a_{p_1} \in B_{\frac{\epsilon}{2}}(\alpha)$.

For chosen $a_{p_1}, \dots, a_{p_{k-1}}$, choose $p_k \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$.

Then, similarly, there is $a_{p_k} \in B_{\frac{\epsilon}{2}}(\alpha)$ such that $p_k > p_{k-1}$.

because no such p_k exists, then it is contradiction to a_{p_k} is a subsequential limit. Therefore, the arbitrary limit point α is contained in $\{a_n\}^*$,

so it is closed.

b). Since a), $\{a_n\}^*$ is closed and bounded, so compact. Thus

$$\sup \{a_n\}^* = \max \{a_n\}^* = \alpha \text{ for some subsequential limit } \alpha.$$

So, given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow a_n < \alpha + \epsilon$ (since $\alpha = \max \{a_n\}^*$)

and there exists infinitely many terms in $\{a_n\}$ such that $\alpha - \epsilon < a_n$ (since $a_{p_k} \rightarrow \alpha$)

Thus $\limsup_{n \rightarrow \infty} a_n = \alpha$

c). Let b_n be defined as:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \dots$$

Then, given $b \in [0, 1]$, there exists a subsequence in $\{b_n\}$ converges to b , because: for any $\epsilon > 0$, there are infinitely many terms in b_n such that

$$|b_n - b| < \epsilon,$$

so we can construct the subsequence. (20)

2025, Fall.

7. Let E be a nonempty bounded subset of a Metric space (X, d) .

10m

Define $\text{diam } E := \sup\{d(x, y) \mid x, y \in E\}$

a). Prove $\text{diam } E = \text{diam } \bar{E}$

b). If E is compact, then $d(x, y) = \text{diam } E$ for some $x, y \in E$.

Proof.

a). Since $E \subseteq \bar{E}$, $\text{diam } E \leq \text{diam } \bar{E}$ is clear.

To show the inverse inequality, using the property of supremum.

Given $\varepsilon > 0$, there exists $x, y \in \bar{E}$ such that

$$\text{diam } \bar{E} - \frac{\varepsilon}{2} < d(x, y) \leq \text{diam } \bar{E}$$

And, since $\bar{E}$ is closure, there are $x', y' \in E$ such that

$$d(x', x) < \frac{\varepsilon}{4} \quad \text{and} \quad d(y, y') < \frac{\varepsilon}{4}.$$

Now, $\text{diam } \bar{E} < d(x, y) + \frac{\varepsilon}{2} < d(x, x') + d(x', y') + d(y', y) + \frac{\varepsilon}{2} < d(x', y') + \varepsilon$.

Since $\varepsilon > 0$ was arbitrary, $\text{diam } \bar{E} \leq \text{diam } E$, so proved. 7m.

b). Since E is compact set, so $E \times E$ is also compact set.

Since the metric function d is continuous, it preserves compactness, so $d[E \times E] \subseteq \mathbb{R}$ is compact set. Therefore,

$$\text{diam } E = \max d[E \times E] \in d[E \times E],$$

so there is $(x', y') \in E \times E$ such that $d(x', y') = \max d[E \times E]$ 3m.

2025 Fall.

8. Let $a > 1$. Evaluate

5m

$$\lim_{t \rightarrow 0^+} \int_t^{at} \frac{\cos x}{x} dx$$

Proof. Consider the Taylor expansion

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

Therefore $\frac{\cos x}{x} = \frac{1}{x} - \frac{x}{2!} + \frac{x^3}{4!} - \dots$

Now,
$$\int_t^{at} \frac{\cos x}{x} dx = \int_t^{at} \frac{1}{x} + \int_t^{at} -\frac{x}{2!} + \frac{x^3}{4!} - \dots dx$$
$$= \ln a + \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx$$

(It is possible, because $\sum (-1)^n x^{2n-1}/(2n)!$ converges for all $x \in \mathbb{R}$)

Now,
$$\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx = 0, \text{ so}$$

the given limit is $\ln a$.

Revised in next

Observation for $\lim_{t \rightarrow 0^+} \int_t^a \frac{\cos x}{x} dx$.

As we know, the power series representation

$$\frac{1}{x} \cdot \cos x = \frac{1}{x} \cdot \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!}$$

is valid for all $x \neq 0$. Therefore, for any $t > 0$,

$$\begin{aligned} \int_t^a \frac{\cos x}{x} dx &= \int_t^a \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} dx \\ &= \int_t^a \frac{1}{x} dx + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \Big|_t^a \end{aligned}$$

Meanwhile, the series converges uniformly in $[t, a]$ for small enough $t > 0$, specifically, $t < \frac{1}{a}$, then for each $n \in \mathbb{N}$,

$$\left| (-1)^n \frac{x^{2n}}{(2n)!} \right| \leq \left| \frac{1}{(2n)!} \right| \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{(2n)!} < \infty,$$

therefore the Weierstrass M-test gives the condition,

Thus, we can exchange the integral to the sum,

$$\int_t^a \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} \int_t^a (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} (-1)^n \left(\frac{1}{2n \cdot (2n)!} \cdot (a^{2n} - t^{2n}) \right)$$

Converges uniformly similarly, now, we can exchange the limit to the sum

$$\begin{aligned} &\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} (-1)^n \frac{1}{2n \cdot (2n)!} \cdot t^{2n} (a^{2n} - 1) \\ &= \sum_{n=1}^{\infty} \lim_{t \rightarrow 0^+} \quad // \\ &= \sum_{n=1}^{\infty} 0 \\ &= 0 \end{aligned}$$

Consequently, the given integral is

$$\int_t^a \frac{1}{x} dx = \ln a - \ln t = \ln a.$$

Note: $\frac{\cos x}{x} = \frac{1}{x} + \frac{\cos x - 1}{x}$, $\left| \int_t^a \frac{\cos x - 1}{x} dx \right| \leq \int_t^a \frac{x}{2} dx \rightarrow 0$ as $t \rightarrow 0$.